



AGROCARE: PLANT DISEASE DETECTION & CROP RECOMMENDATION SYSTEM

A PROJECT REPORT

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Submitted by

Name of Student

UPAYAN CHAKRABORTY

SANDIPAN RAKSHIT

PRERANA MAITI

MRIGANGANA SARKAR

SOUVIK GHOSH

Examination Roll No.

11600222134

11600222092

11600222070

11600222062

11600222117

Under the supervision of

Prof. (Dr.) Subro S. Thakur

Department of Computer Science & Engineering,

MCKV Institute of Engineering, 243,

G.T. Road(N), Liluah, Howrah – 711204

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
MCKV Institute of Engineering
An Autonomous Institute Affiliated to
Maulana Abul Kalam Azad University of Technology, West Bengal

CERTIFICATE

This is to certify that the project report entitled “AGROCARE: PLANT DISEASE DETECTION & CROP RECOMMENDATION SYSTEM” submitted by “UPAYAN CHAKRABORTY, SANDIPAN RAKSHIT, PRERANA MAITI, SOUVIK GHOSH, MRIGANGANA SARKAR” for 8th semester examination has been prepared for the partial fulfillment of the degree of Bachelor of Technology in Computer Science and Engineering, awarded by the Maulana Abul Kalam Azad University of Technology, West Bengal. They have carried out the project work under my supervision.

HEAD OF THE DEPARTMENT

Mr. Avijit Bose
Assistant Professor
Dept. of Computer Science and Engg.
MCKV Institute of Engineering
243, G.T. Road (North),
Liluah Howrah – 711204

SUPERVISOR

Prof. (Dr.) Subro S. Thakur
Professor
Dept. of Computer Science and Engg.
MCKV Institute of Engineering
243, G.T. Road (North),
Liluah Howrah – 711204

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
MCKV Institute of Engineering
An Autonomous Institute Affiliated to
Maulana Abul Kalam Azad University of Technology, West Bengal

CERTIFICATE OF APPROVAL
(B.Tech Degree in Computer Science and Engineering)

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SIGNATURE OF EXAMINATION COMMITTEE MEMBERS FOR EVALUATION OF PROJECT REPORT	1.
	2.
	3.
	4.
	5.

Date of evaluation:

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1. **Upayan Chakraborty** _____
2. **Sandipan Rakshit** _____
3. **Prerana Maiti** _____
4. **Mrigangana Sarkar** _____
5. **Souvik Ghosh** _____

ABSTRACT

Agriculture remains the backbone of food security and economic stability, especially in developing regions where farming supports the majority of livelihoods. To address challenges such as crop diseases, unpredictable weather, and inefficient resource use, AgroCare was developed as an AI-driven agricultural assistance system integrating Plant Disease Detection and Crop Recommendation modules. The system combines deep learning, machine learning, and IoT technologies to deliver real-time, data-driven insights that enhance productivity and promote sustainable farming practices.

The Crop Recommendation System is an intelligent agriculture-based solution that integrates hardware sensors, machine learning, and a conversational chatbot to assist farmers in selecting the most suitable crop for cultivation. The system collects real-time soil parameters such as soil moisture, temperature, pH, and essential nutrients (Nitrogen, Phosphorus, and Potassium) using sensors interfaced with an Arduino Nano. Additionally, environmental data including temperature, humidity, and rainfall is obtained through external APIs.

All collected data is processed and provided as input to a machine learning model, which analyzes the conditions and predicts the most appropriate crop for a given environment. The predicted result is displayed through a user-friendly web interface.

To further enhance usability, a chatbot powered by the Gemini API is integrated into the system. This chatbot enables users to interact using natural language queries and provides structured, agriculture-related guidance such as crop selection, soil management, and fertilizer recommendations.

The system combines IoT, machine learning, and natural language processing to create a smart, interactive, and data-driven agricultural support platform, helping improve productivity and decision-making in farming.

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1. INTRODUCTION

Agriculture plays a vital role in the economy, and selecting the right crop based on soil and environmental conditions is crucial for maximizing yield and maintaining soil health. Traditional farming methods often rely on experience and manual observation, which may lead to inaccurate decisions due to changing environmental factors.

With advancements in technology, smart agriculture solutions have emerged that utilize Internet of Things (IoT), machine learning, and data analytics to provide precise and real-time insights. This project presents a Crop Recommendation System that leverages these technologies to assist farmers in making informed decisions.

The system uses multiple soil sensors, including soil moisture, temperature, pH, and NPK sensors, connected to an Arduino Nano to collect real-time data from the field. Along with this, environmental parameters such as temperature, humidity, and rainfall are obtained using online APIs. These inputs are then processed by a machine learning model to predict the most suitable crop for cultivation.

In addition to prediction, the system includes an interactive chatbot integrated using the Gemini API. The chatbot allows users to ask questions related to crops, soil conditions, and farming practices, providing instant and structured responses.

The combination of hardware sensing, intelligent prediction, and conversational assistance makes this system a comprehensive and user-friendly solution for modern agriculture. It aims to improve crop productivity, optimize resource usage, and support farmers with accurate and real-time recommendations.

2. LITERATURE REVIEW

In recent years, artificial intelligence and data-driven techniques have gained significant importance in agriculture, with numerous studies focusing on plant disease detection and crop recommendation. These systems aim to enhance productivity, reduce crop losses, and support farmers with accurate, data-based decision-making.

Early plant disease detection research relied on traditional machine learning methods such as SVM and Random Forest, using manually extracted features from leaf images (Patil et al., 2019; Singh et al., 2020). However, these models showed limited accuracy under varying environmental conditions. With advancements in deep learning, Convolutional Neural Networks (CNNs) demonstrated superior performance by automatically learning features and achieving high accuracy in disease classification, as highlighted by Ferentinos (2018) with results exceeding 95%.

For crop recommendation, earlier studies like Akinbile et al. (2021) utilized soil parameters (NPK, pH) to suggest suitable crops, while others incorporated climatic factors such as temperature and rainfall for better reliability. IoT-based systems proposed by Rajeswari et al. (2022) enabled real-time soil monitoring but lacked integration of external weather data, limiting region-specific adaptability.

A review of existing research indicates that most solutions address either plant disease detection or crop recommendation independently. Only a few combine both, and even fewer integrate real-time soil sensing hardware with dynamic weather data from APIs. The absence of such hybrid systems creates a gap in achieving a more comprehensive and adaptive agricultural support tool.

To address these limitations, the proposed project integrates hardware-based soil sensing with API-driven environmental data to enhance both plant disease detection and crop recommendation. By combining real-time soil information with dynamic temperature, humidity, and rainfall inputs, the system is designed to deliver a more accurate, automated, and reliable smart agriculture solution capable of supporting farmers across diverse agricultural conditions.

3. SCHEME AND IMPLEMENTATION

3.1 System Architecture Overview

The Crop Recommendation System is designed using an integrated architecture that combines hardware sensors, software processing, machine learning, and a chatbot interface to provide accurate crop suggestions.

The system begins with the input layer, where soil parameters are collected using sensors such as soil moisture, soil temperature, pH sensor, and NPK sensor. These sensors are connected to an Arduino Nano, which reads both analogue and digital signals and displays the data on the Serial Monitor. In addition to sensor data, environmental parameters like temperature, humidity, and rainfall are fetched using the OpenWeatherMap API.

In the processing layer, the Arduino acts as the data acquisition unit, while the frontend interface allows users to manually input sensor readings. This data is sent to the backend server, where input validation and preprocessing are performed, such as checking for empty fields, removing extra spaces, and formatting the data.

The processed data is then passed to the machine learning layer, where a trained classification model analyses parameters like N, P, K values, pH, soil moisture, soil temperature, humidity, rainfall, and temperature. Based on these inputs, the model predicts the most suitable crop for cultivation.

In the application layer, a user-friendly web interface displays input fields and prediction results. The predicted crop is shown to the user along with a descriptive message. Additionally, a chatbot integrated using the Gemini API allows users to interact with the system using natural language queries related to crops, fertilizers, and soil conditions.

The chatbot implementation involves steps such as input validation, preprocessing (lowercase conversion and keyword detection), context handling, and system prompt injection. The query is then processed by the large language model, which uses tokenization, embeddings, transformer layers, and Softmax functions to generate meaningful responses.

The hardware implementation includes proper sensor interfacing with Arduino Nano, RS485 communication for the NPK sensor, and a regulated power supply using an LM7805 voltage regulator to convert 12V to 5V. Capacitors are used to ensure voltage stability and noise reduction.

This integrated implementation ensures accurate, real-time crop prediction and provides an intelligent, interactive solution for modern agriculture.

4. PREVIOUS WORKS

4.1 Overview

- In the previous phase of the project (seventh semester), the AgroCare system was developed as a foundational prototype focusing on two primary modules: Plant Disease Detection and Crop Recommendation.
- The Plant Disease Detection module was implemented using basic Convolutional Neural Network (CNN) models trained on publicly available datasets. The system was capable of classifying plant diseases from leaf images with reasonable accuracy, but it lacked optimization in terms of model performance, scalability, and dataset diversity.
- The Crop Recommendation System was developed using machine learning algorithms trained on the Kaggle Crop Recommendation Dataset. The model utilized key parameters such as Nitrogen (N), Phosphorus (P), Potassium (K), pH, temperature, humidity, and rainfall to predict suitable crops. A web interface was created using Flask, where users could manually input these parameters to obtain recommendations.
- On the hardware side, an initial prototype was built using Arduino Nano integrated with basic sensors such as soil moisture and soil temperature sensors. The sensor data was displayed on the serial monitor, and users were required to manually transfer these values into the web application for further processing.
- However, the system had certain limitations. The hardware setup was incomplete, as advanced sensors like NPK and pH were not fully integrated. There was no real-time automation between hardware and software components, and the system relied heavily on manual data entry. Additionally, external environmental data was not dynamically fetched, which limited the accuracy and adaptability of the recommendations.
- These limitations highlighted the need for improved system integration, real-time data

acquisition, and enhanced model performance, which have been addressed in the current phase of the project.

Table 1. Key Variables and their Importance

Variable	Description
Nitrogen (N)	Essential for leaf growth and chlorophyll formation in plants.
Phosphorus (P)	Promotes root development and improves flowering and fruiting.
Potassium (K)	Enhances overall plant health and disease resistance.
Temperature	Affects seed germination, growth rate, and crop yield.
Humidity	Influences transpiration rate and water availability to plants.
pH	Determines soil acidity or alkalinity, affecting nutrient absorption.
Rainfall	Provides necessary water for crop growth and soil moisture balance.
Soil Temperature	Impacts microbial activity and nutrient availability in the soil.
Soil Moisture	Maintains hydration levels crucial for seed germination and plant growth.

4.2 Use Case Scenario:

In this system, hardware sensors provide real-time soil data such as soil temperature, soil moisture, and pH, while an API retrieves weather parameters including rainfall, humidity, and temperature. Along with soil nutrient values (N, P, K), these inputs are processed by the machine learning model to recommend the most suitable crop for the given conditions. This combination of hardware sensing and software prediction ensures accurate, location-specific recommendations, enabling farmers to make informed and efficient agricultural decisions.

Model Info:

- **Type:** Supervised classification
- **Algorithms Used:** Decision Tree, Random Forest, KNN, XGBoost
- **Input:** 9 numerical features
- **Output:** Predicted crop label (e.g., rice, wheat, maize, muskmelon, etc.)

5. WORKFLOW DIAGRAM

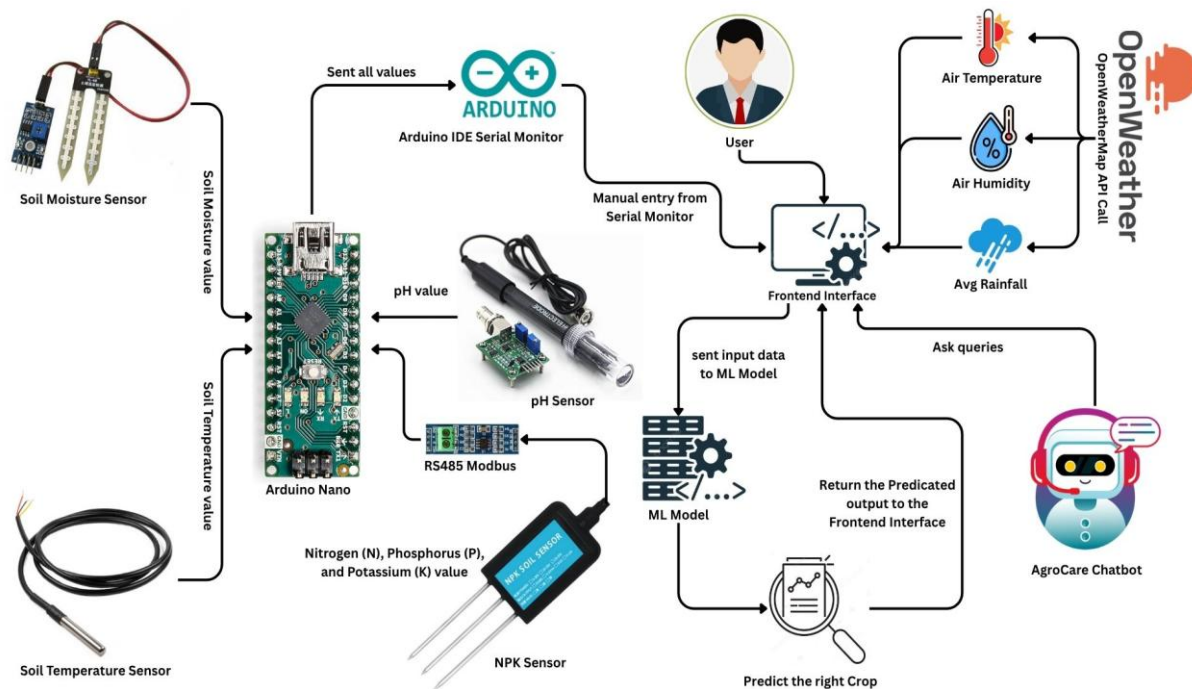


Fig 1. Workflow diagram

The flow diagram (Fig 1.) illustrates how different components work together in the Smart Crop Prediction System.

5.1 Soil Sensors (Input Stage):

a. Soil Moisture Sensor

The soil moisture sensor measures the water content present in the soil by analyzing its electrical resistance. Wet soil has low resistance and higher conductivity, while dry soil has high resistance. The sensor sends analog or digital signals to the Arduino Nano, which helps in determining irrigation needs and maintaining proper soil moisture levels.

b. Soil Temperature Sensor

The soil temperature sensor measures the temperature of the soil, which is important for plant growth, seed germination, and microbial activity. Sensors like DS18B20 provide digital output directly to the Arduino Nano. This data helps in monitoring environmental conditions and ensuring suitable temperature for crops.

c. pH Sensor

The pH sensor is used to measure the acidity or alkalinity of the soil on a scale of 0 to 14. It uses an electrode probe to detect hydrogen ion concentration and produces an analog output signal. The Arduino Nano processes this data, which helps in selecting appropriate crops and managing soil health.

d. NPK Sensor

The NPK sensor measures essential soil nutrients such as Nitrogen (N), Phosphorus (P), and Potassium (K), which are crucial for plant growth and development. It works using ion-sensing techniques and communicates with the Arduino Nano through the RS485 Modbus protocol. The collected data is useful for analyzing soil fertility and recommending fertilizers.

5.2 Arduino Processing**(a) Reading and Displaying Sensor Data**

The Arduino Nano reads real-time data from all connected soil sensors, including soil moisture, temperature, pH, and NPK values. It processes these inputs and displays the collected data on the Arduino IDE Serial Monitor, allowing users to monitor the sensor readings continuously.

(b) Data Transmission to Frontend Interface

After processing, the Arduino Nano sends the collected soil parameters to the frontend interface. This data is used for further analysis, visualization, and integration with advanced systems such as machine learning models for crop prediction.

5.3 Frontend Interface:

- The user interacts with the system through this interface.
- It collects three types of inputs:
 - **From Arduino:** soil moisture, temperature, pH, and NPK values manually entered the output from the Arduino serial monitor.
 - **From OpenWeatherMap API:** Air temperature, humidity, and average rainfall.

5.4 Data Flow to Machine Learning Model:

- All the collected input values (sensor data, API data, and manual entries) are sent to the ML Model.
- The model processes these inputs to predict the most suitable crop for the given conditions.

5.5 Prediction Output:

- The predicted crop is sent back from the ML Model to the Frontend Interface.
- The result is displayed to the user, completing the cycle.

6. EXPERIMENTAL SETUP

The Crop Recommendation System combines both hardware and software components to predict the most suitable crop for cultivation based on soil and environmental data.

6.1 SOFTWARE SETUP

6.1.1 Chatbot Integration using Gemini API

The system was further enhanced by integrating a chatbot using the Gemini API. This chatbot enables users to interact with the system using natural language queries.

It is connected with the backend and can access real-time sensor data and prediction outputs. Users can ask questions related to crop recommendations, soil conditions, and fertilizers, and receive intelligent responses instantly. This integration improves user experience by making the system more interactive and user-friendly.

6.1.2 AgroCare Chatbot Workflow Diagram

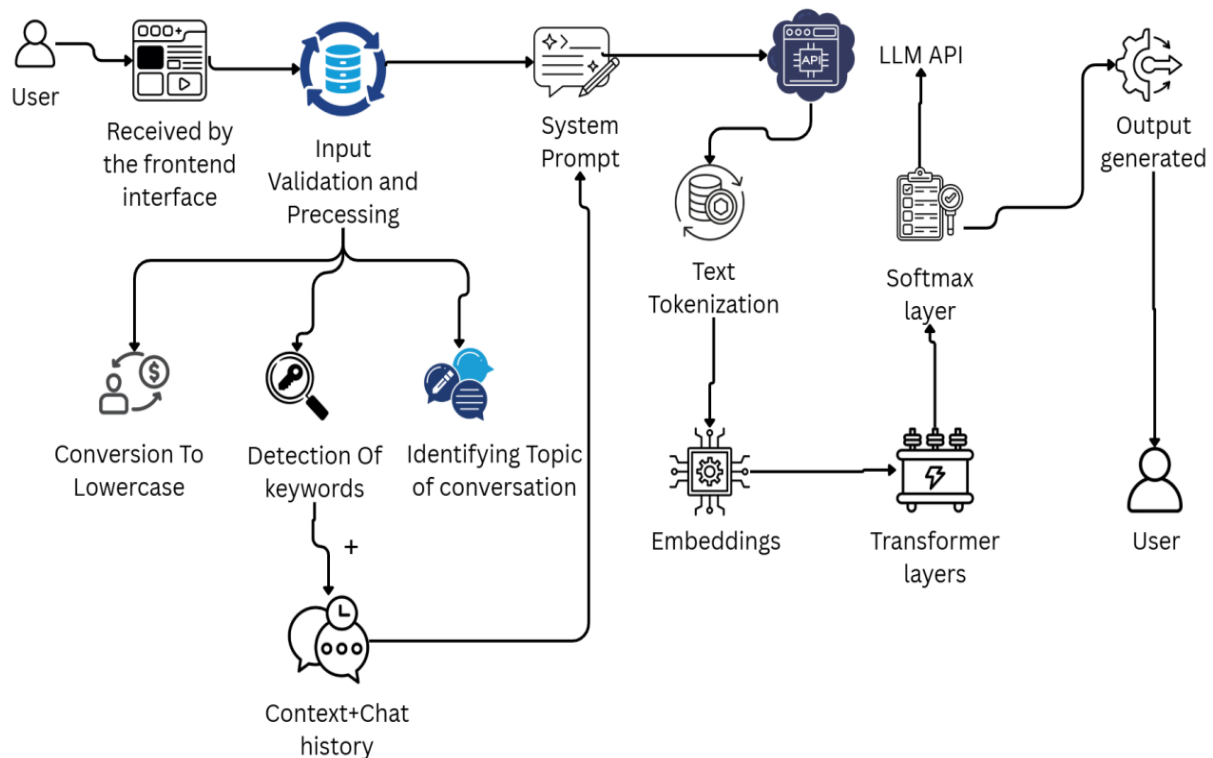


Fig 2.1 Work Flow Diagram of Chatbot

6.1.3 User Interaction Through Frontend Interface

The first step in the process of the chatbot system is when the chatbot gets accessed by the user via web/mobile application. User inputs the query concerning agriculture, which may be about the disease of crops, how to treat such issues, fertilizer, or what type of plants will be good for particular types of soils. The communication process between the user and the chatbot happens with the use of a chat-based user interface, consisting of two elements –

- Input field for typing the messages
- Display field for displaying the response to users' requests.

Frontend technologies, including HTML, CSS, and JavaScript, are utilized during developing responsive UIs.

6.1.4 Input Validation and Preprocessing

Once the user's message reaches the backend API, the system performs input validation to ensure the request contains a valid query before further processing.

The system checks:

- Whether the message field exists in the request
- Whether the message is empty
- Removes extra spaces from the beginning and end of the message

After these checks, the cleaned message is processed further by the chatbot. If the message is missing or empty, the system immediately returns an error response and stops further processing, ensuring that invalid inputs are not sent to the AI model.

6.1.5 Processing of User Input

After the input is validated, the chatbot performs preprocessing on the user's text query to make it easier for the system to analyse and interpret the message correctly.

Preprocessing includes:

- Converting the message to lowercase
- Detecting keywords related to crops
- Identifying the topic of the conversation

6.1.6 Context and Chat History handling

The chatbot identifies the main topic of the user's query by detecting crop-related keywords in the message, such as rice, coffee, wheat, maize, cotton, banana, and mango. If any of these keywords are found, the system updates the current conversation topic, this helps the chatbot maintain contextual understanding and respond accurately to follow-up questions related to the same crop.

6.1.7 System Prompt Injection

Before sending the user query to the AI model, the system adds a predefined system prompt that guides how the chatbot should generate responses. This prompt defines the role and

response style of the chatbot so that the answers remain relevant to agriculture and are presented in a clear format.

The system prompt instructs the chatbot to:

- Act as an Agricultural Expert Assistant
- Provide Short Structured Bullet Points
- Give Step-by-Step Practical Solutions
- Focus on Crop and Farming Advice

6.1.8 LLM Integration

6.1.8.1 Text Tokenization

Text tokenization is the process of breaking the input sentence into smaller units called **tokens**. These tokens are then converted into numerical identifiers so that the neural network inside the LLM can process the input text.

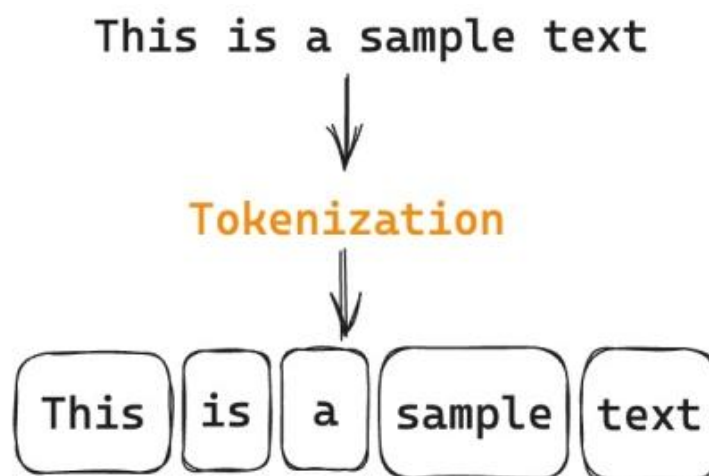


Fig 2.2. Text Tokenization

6.1.8.2 Embedding Creation

Vector Embedding: Vector embedding converts tokens into numerical vectors so the neural network can understand their meaning. These vectors help the model capture relationships between different words.

Positional Embedding: Positional embedding provides information about the position of each token in a sentence. This helps the model understand the correct order of words in the input text.

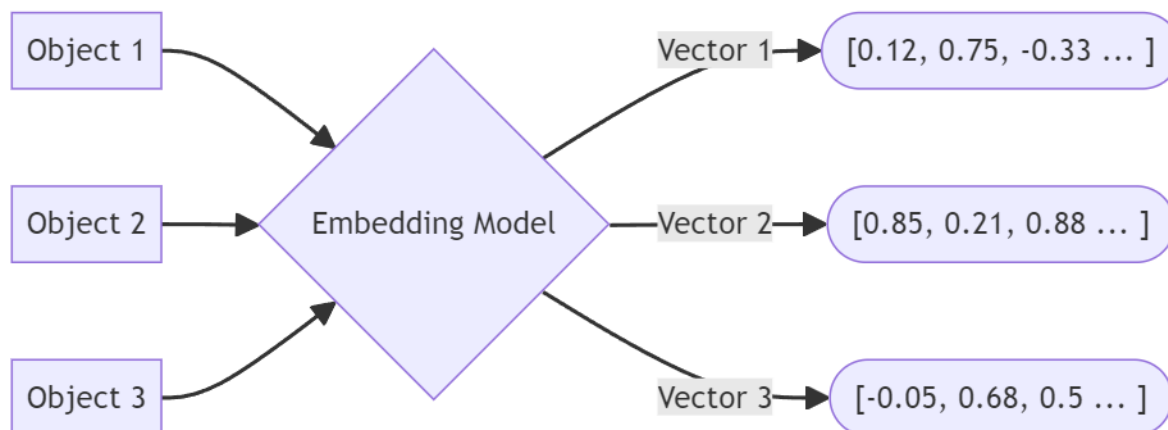


Fig 2.3. Embedding Model

6.1.8.3 Transformer Layers

Transformer layers process the input embeddings to understand the relationships between words in a sentence. They use mechanisms like self-attention and feed-forward networks to capture context and generate meaningful outputs.

- **Self-Attention Mechanism** – Helps the model understand relationships between different words in a sentence.
- **Multi-Head Attention** – Allows the model to focus on multiple parts of the sentence simultaneously.
- **Feed-Forward Neural Network** – Processes the attention output and transforms it into deeper representations.
- **Layer Normalization** – Stabilizes and speeds up the training of the model.
- **Residual Connections** – Helps maintain information flow across layers and prevents information loss.

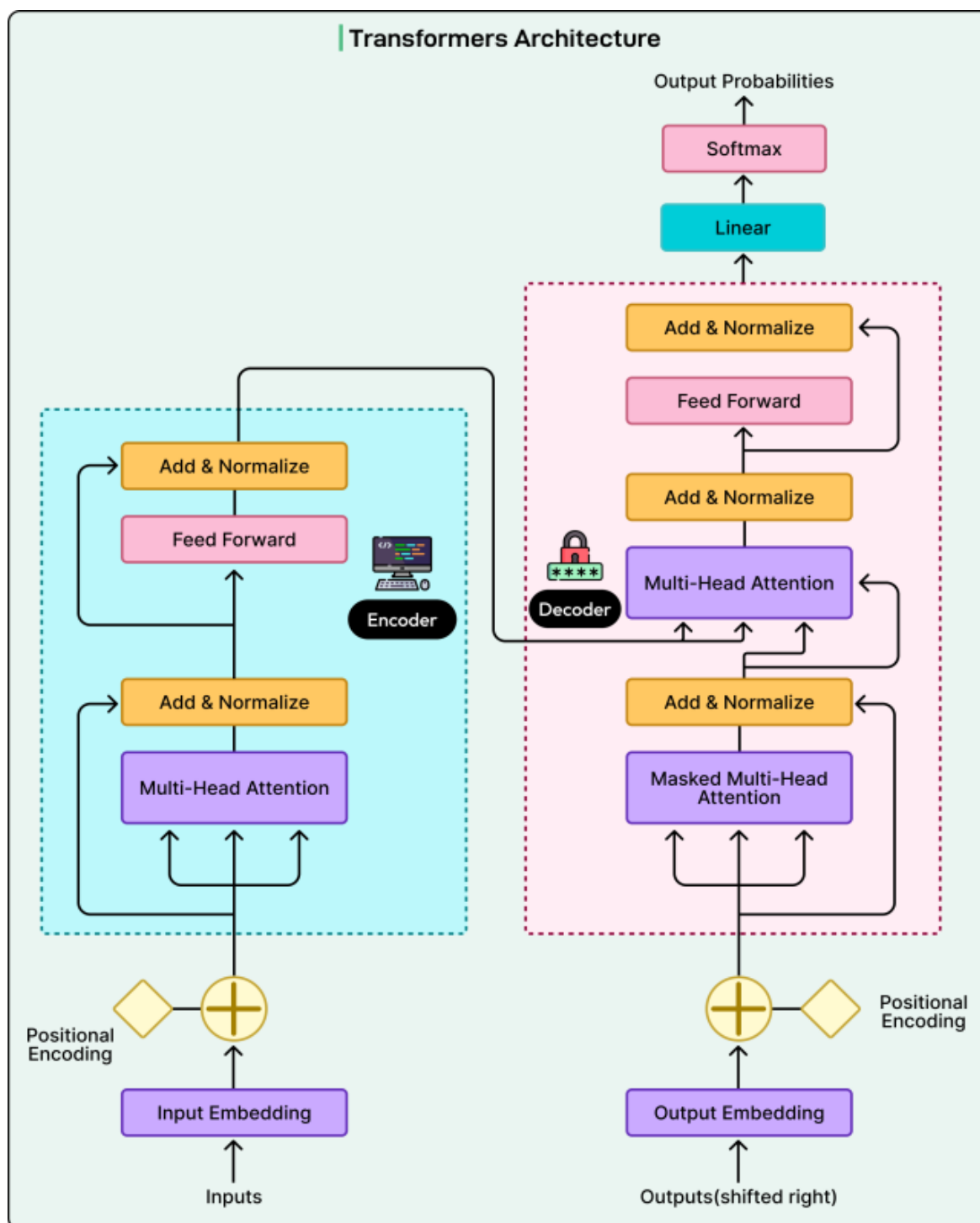


Fig 2.4. Transformer layers

6.1.8.4 SoftMax Layer

After processing the input through transformer layers, the model uses a Softmax layer to calculate the probability of possible next words. The word with the highest probability is selected as the next token, and this process repeats until a complete response is generated.

6.1.8.5 Output Generation (Final Response Generation)

After the model predicts the sequence of tokens, these tokens are converted back into readable text to form the final response. The generated response is then sent from the backend server to the frontend chatbot interface. Finally, the answer is displayed to the user in the chat window, completing the interaction cycle.

6.2 HARDWARE SETUP

6.2.1 7th Semester Work Summary

In the 7th semester, the initial hardware setup was developed by integrating the soil moisture sensor and DS18B20 soil temperature sensor with the Arduino Nano microcontroller. The connections were implemented using a breadboard, forming a temporary prototype setup. This arrangement allowed real-time collection of basic soil parameters such as moisture level and temperature for testing and validation of the system.

6.2.2 NPK Sensor Integration

The system was further enhanced by integrating the NPK sensor with the Arduino Nano using an RS485 communication module. This enabled the measurement of essential soil nutrients such as Nitrogen (N), Phosphorus (P), and Potassium (K). The integration allowed the system to analyze soil fertility more effectively and provide better crop recommendations.

6.2.3 Soil pH Sensor Integration

A soil pH sensor was added to the system to measure the acidity or alkalinity of the soil. It was connected to the Arduino Nano using analog input pins, allowing real-time monitoring of pH values. This addition helped in improving the accuracy of crop prediction by considering soil chemical properties.

6.2.4 CIRCUIT DIAGRAM:

6.2.4.1 Power Supply Design

The power supply circuit is designed to provide two different voltage levels required by the system: +12V for the NPK sensor and +5V for the soil moisture, soil temperature, and pH sensors.

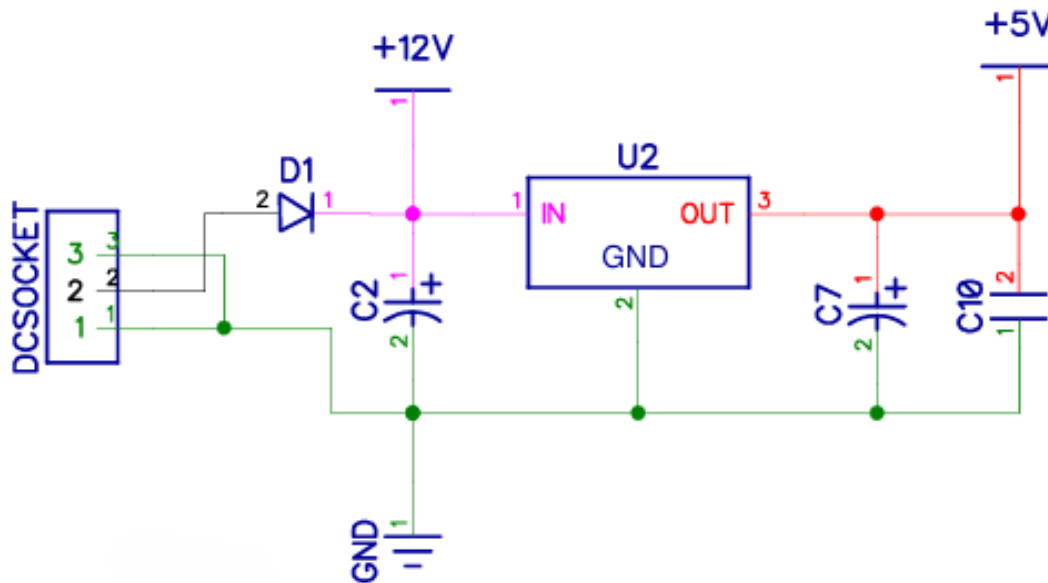


Fig 3. Power supply diagram

6.2.4.1.1 Power Source and 12V Supply

A 12V, 1A DC adapter is used as the main power source. This adapter converts AC mains supply into a regulated +12V DC output.

- The +12V line is directly connected to the NPK sensor, as it requires higher operating voltage.
- The ground (GND) is shared across all components to maintain a common reference.

6.2.4.1.2 12V to 5V Conversion Circuit

To supply power to low-voltage sensors, the +12V input is stepped down to +5V using a voltage regulator circuit built around the LM7805 regulator IC.

Components Used:

- LM7805 Voltage Regulator
- C2 = 100 μ F (input filter capacitor)
- C7 = 47 μ F (output filter capacitor)
- C10 = 0.1 μ F (noise filtering capacitor)
- Heat sink (to dissipate excess heat from LM7805)

Working:

- The +12V input is first passed through the C2 (100 μ F capacitor) to smooth any fluctuations.
- This filtered voltage is fed into the input pin of LM7805.
- The LM7805 regulates the voltage and provides a stable +5V output.
- At the output:
 - C7 (47 μ F) is used for smoothing and stabilizing the output voltage.
 - C10 (0.1 μ F) is used to eliminate high-frequency noise.
- A heat sink is attached to the LM7805 to prevent overheating during operation.

6.2.4.1.3 Distribution of Power

- +12V Line → NPK Sensor
- +5V Line → Soil Moisture Sensor, Soil Temperature Sensor, pH Sensor, and Arduino Nano
- Common GND → All components

This ensures that each component receives the required voltage level safely and efficiently.

6.2.4.2 Sensor and Arduino Integration

The circuit is designed to interface multiple soil sensors with the Arduino Nano, enabling the acquisition of real-time soil parameters such as moisture content, temperature, pH level, and essential nutrient concentrations (Nitrogen, Phosphorus, and Potassium). The Arduino Nano acts as the central processing unit, collecting both analog and digital signals from the sensors and preparing them for further analysis.

To ensure proper operation of all components, the system utilizes a dual power supply configuration consisting of +12V and +5V lines. The +12V supply is directly used to power high-voltage components such as the NPK sensor, while a regulated +5V supply—derived using a voltage regulator circuit—is used for the Arduino Nano and other low-power sensors. A common ground (GND) is maintained across the entire circuit to provide a stable reference point and ensure reliable communication between all interconnected components.

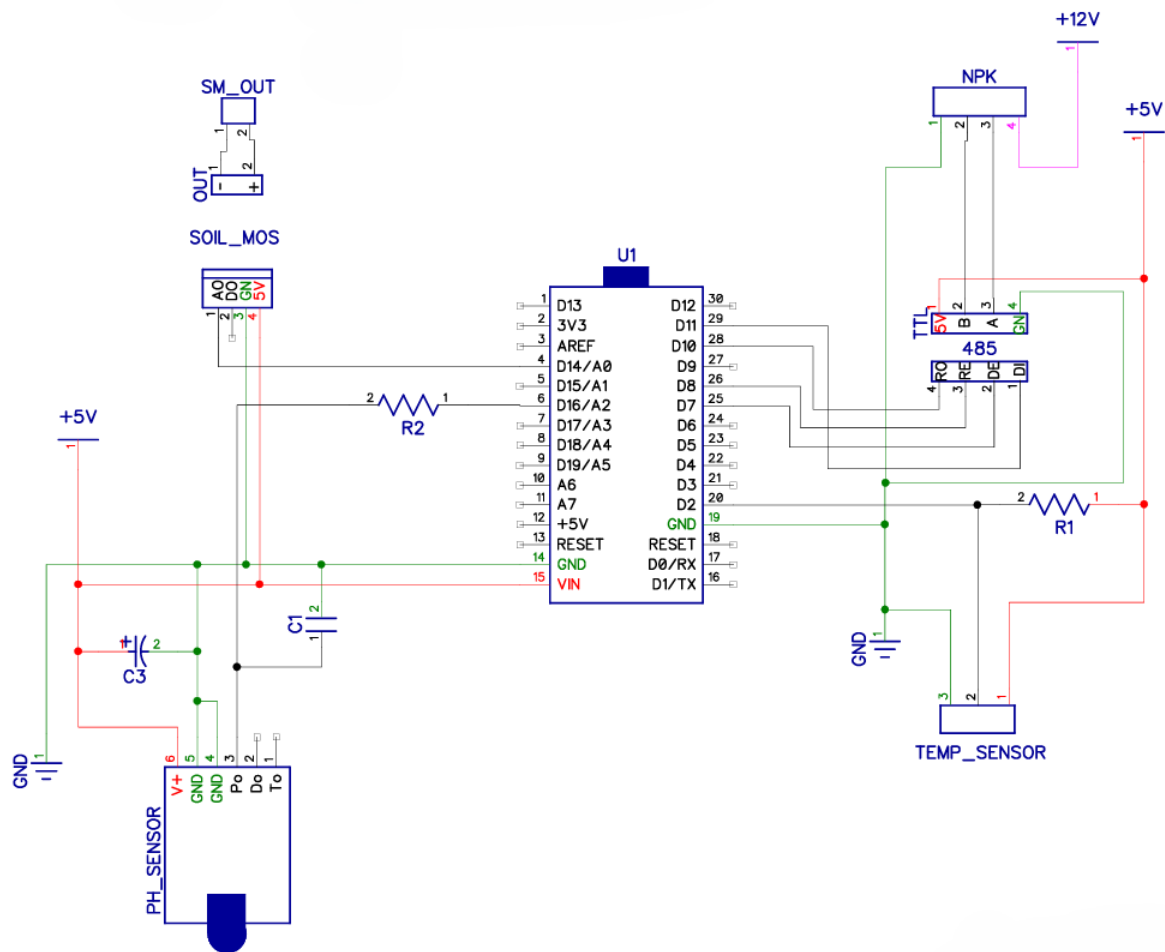


Fig 4. Circuit diagram

6.2.4.2.1 Arduino Nano

The Arduino Nano acts as the central controller, interfacing with all sensors and communication modules. It reads analog and digital signals and processes them for further transmission.

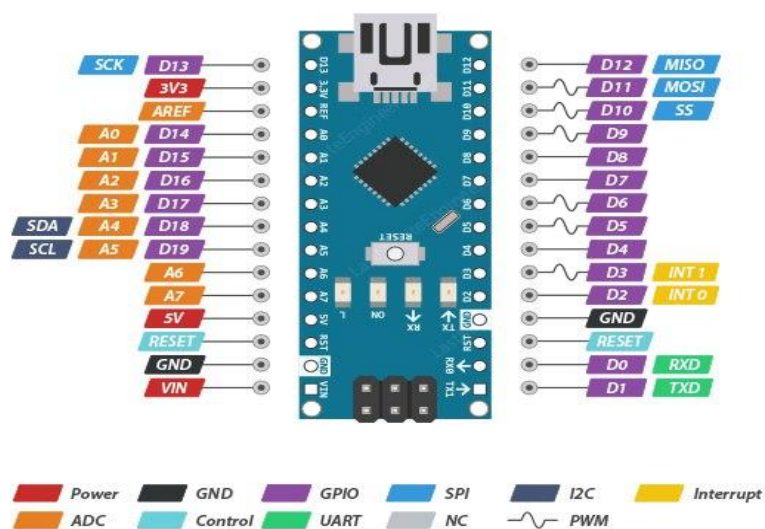


Fig 5.1. Arduino Nano

6.2.4.2.2 Soil Moisture Sensor Connection

The Soil Moisture Sensor (Fig 4.) measures the water content in the soil to determine how wet or dry it is. It works by detecting changes in electrical resistance or capacitance between its probes. This helps in monitoring irrigation needs and maintaining optimal soil conditions for crop growth.

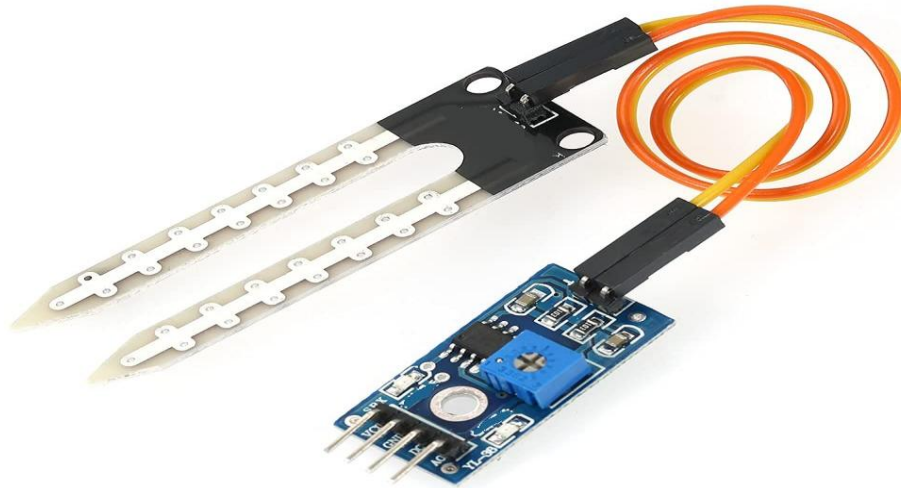


Fig 5.2. Soil Moisture Sensor

Table 2. Soil Moisture Sensor to Arduino Nano Connection

Soil Moisture Sensor pin	Arduino Nano Pin
GND	GND
VCC	3V3
A0	A0

6.2.4.2.3 Soil Temperature Sensor (DS18B20) Connection

The DS18B20 (Fig 3.) is a digital temperature sensor that provides accurate temperature readings from -55°C to $+125^{\circ}\text{C}$. It uses a 1-Wire communication protocol, allowing multiple sensors to connect through a single data pin. The sensor is ideal for soil and environmental temperature monitoring in agricultural applications due to its high precision ($\pm 0.5^{\circ}\text{C}$) and easy digital interface.



Fig 5.3. DS18B20 Temperature Sensor

Table 3. Temperature Sensor to Arduino Nano Connection

DS18B20 Pin	Arduino Nano
GND	GND
DATA	D2
VCC	5V or 3.3V

Table 4. Additional Components connected (Temperature Sensor)

Component	Value	Connection	Purpose
R1	4.7 k Ω	Between DATA pin and +5V	Acts as pull-up resistor to keep data line HIGH and enable stable 1-Wire communication

6.2.4.2.4 Soil pH Sensor Connection

The soil pH sensor module (commonly based on PH-4502C) is interfaced with the Arduino Nano to measure soil acidity or alkalinity.



Fig 5.4. Soil pH Sensor

Table 5. pH Sensor to Arduino Nano Connection

pH Sensor Pin	Arduino Nano Pin
VCC (Pin 6)	5V
GND (Pin 5)	GND
PO (Pin 3, Analog Output)	A0

The PO pin provides an analog voltage corresponding to the pH value, which is read by the Arduino through its analog input pin.

Table 6. Additional Components connected (pH Sensor)

Component	Value	Connection	Purpose
R2	1 k Ω	Between pH sensor output (PO) and Arduino analog pin (A0)	Signal conditioning and input protection
C1	0.1 μ F	Between signal line and GND	Filters high-frequency noise and stabilizes signal
C3	10 μ F	Between +5V and GND	Smooths power supply and reduces voltage fluctuations

6.2.4.2.5 NPK Sensor (JXBS-3001) Connection

The NPK (JXBS-3001) sensor is interfaced with the Arduino Nano using an RS485 communication module, as it operates on Modbus protocol and requires higher voltage.

**Fig 5.5. NPK Sensor**

Power Supply Connection:

- **VCC** → +12V (Direct from power supply)
- **GND** → GND (Common ground)

The NPK sensor requires a direct +12V supply for proper operation, which is provided from the main power source without voltage regulation.

Table 7. NPK Sensor Connection to RS485 Module

Wire Colour of NPK Sensor	Connection to RS485 Module
Red (VCC)	12V
Black (GND)	GND
Yellow (A)	RS485-A
Blue (B)	RS485-B

Table 8. RS485 Module to Arduino Nano Connection

RS485 Module Pin	Arduino Nano Pin
RO	D10
DI	D11
RE	D8
DE	D7
VCC	5V
GND	GND

6.2.4.3 PCB Design and Implementation

6.2.4.3.1 Overview of PCB Design

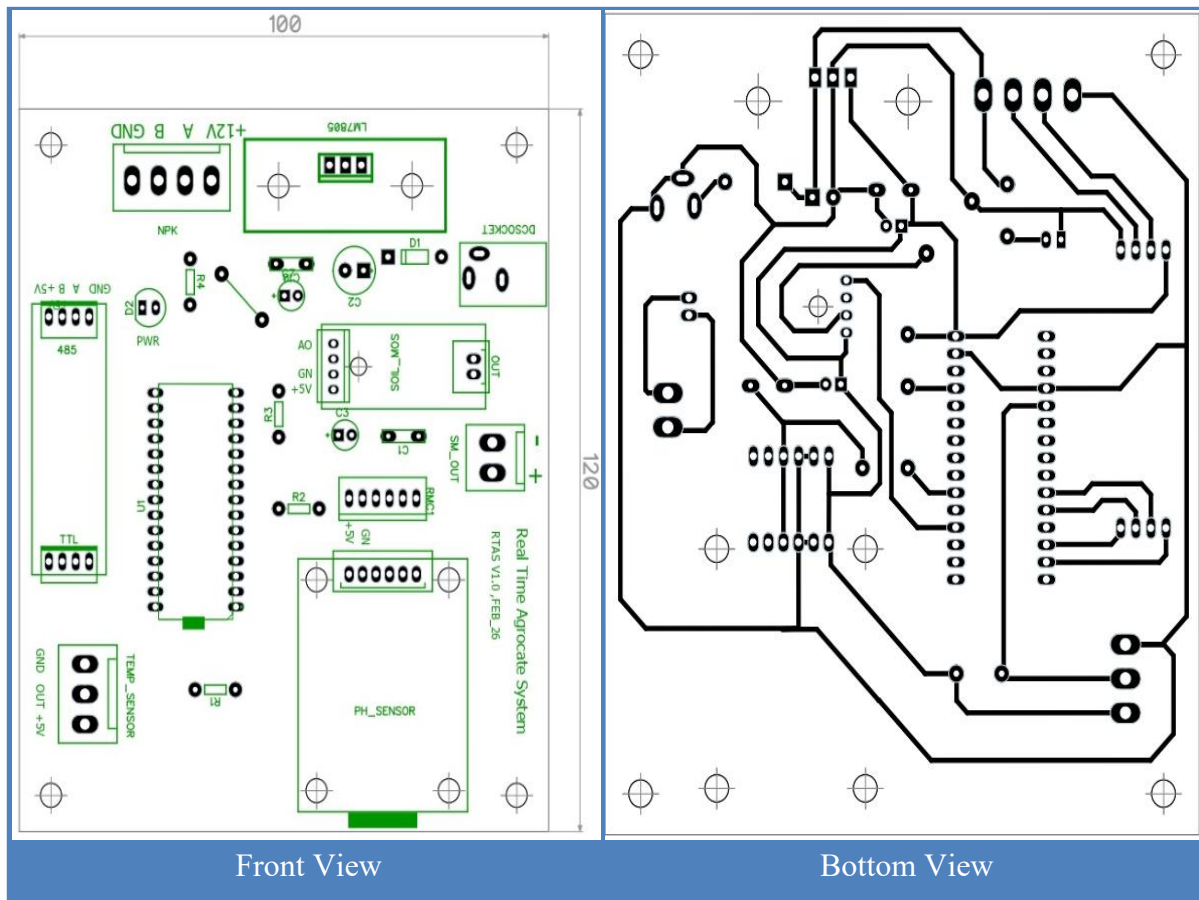


Fig 6. PCB Design

The project utilizes a custom-designed, single-layer PCB to integrate various environmental sensors with a central processing unit. The board (Fig.3) serves as the hardware backbone for data acquisition and communication.

1. Power Management Circuitry

To ensure stable operation, the PCB features a dedicated power regulation stage:

- **Input Voltage:** The board accepts power through a standard DC Socket.
- **Regulation:** An LM7805 linear voltage regulator is used to step down the input voltage to a stable +5V required by the microcontroller and sensors.
- **Filtering & Indicators:** Electrolytic capacitors (C1, C2) are implemented for noise decoupling, and a Power LED (PWR) provides a visual status of the board's power state.

2. Core Processing & Connectivity

The central component of the board is a 30-pin microcontroller socket (U1), likely designed for an Arduino Nano or similar footprint. For data transmission, the board includes:

- **RS485 Interface:** A dedicated section for a 485 module allows for long-distance, robust serial communication using the GND, A, B, +5V pinout.
- **TTL Header:** A TTL interface is provided for direct serial debugging or low-level communication.

3. Sensor Integration & Interfaces

The PCB is designed to interface with multiple agricultural sensors through specialized headers:

- **Soil Moisture:** A header labeled SOIL_MOS monitors ground water content, with a corresponding SM_OUT for signal processing.
- **Temperature:** A dedicated TEMP_SENSOR port provides GND, OUT, and +5V pins.
- **pH Monitoring:** A large footprint is reserved for a PH_SENSOR module, essential for soil health analysis.
- **NPK Interface:** A header labeled NPK is included, facilitating connection to sensors that measure Nitrogen, Phosphorus, and Potassium levels.

4. Physical Construction

- **Layout:** The Bottom Copper Layer shows a clear routing strategy with thickened traces for power rails to handle current loads and isolated signal paths to minimize interference.
- **Dimensions:** The board measures approximately 120mm in length.
- **Mounting:** Four corner mounting holes are provided for secure installation within a protective enclosure.

6.2.4.3.2 PCB Connections Review

In contrast to the breadboard, the PCB utilizes fixed copper tracks and dedicated connectors instead of jumper wires. Silkscreen printing designates every connection point, for example, +12V, GND, A, B, etc. The power input for the PCB comes from a three-pin screw-terminal block (50/3), designated +12 V and GND, with an option for the inclusion of an integrated

switch or shielding. Two more two-pin screw-terminal blocks (50/2) offer extra connections, e.g., the differential bus or sensor power connections. The reason for using screw terminal blocks is the design with screw clamp contacts, allowing compact high pressure low resistance contacts capable of holding wires safely despite vibrations and large currents.

Also, the PCB uses 2.54 mm pin headers called Berg Stick connectors. They include female header strips and male header strip. Female headers allow connecting sensor cable or a jumper lead directly to the sensor. For example, a 3-pin female header may be silkscreen labeled as “A0, GND, +5V,” providing analog input from the sensor along with its power and ground connection. The male header offers a connector to plug the PCB into a microcontroller. Such pin headers enable plugging connections without soldering, simplifying assembly and debugging process. In conclusion, the traces on the PCB carry the same signal and power as in the breadboard, but every signal or voltage line is connected through a soldered connector or header, providing a strong connection.

6.2.4.3.3 Components and Their Purposes

- **3-pin PCB Mount Terminal Block (50/3)** - A screw terminal, used as the primary input for power and may be used for the inclusion of a switch or common ground point, consists of three posts screwed in place (e.g., +12V, GND, and additional post). Conductor leads can be securely tightened to each of the posts. Since screw terminals allow greater conductor gauge and currents, their high contact pressure reduces contact resistance, ensuring reliable power supply.



Fig 7.1 3-pin PCB Mount Terminal Block

- **2-pin PCB Mount Terminal Block (50/2)** – Two equal-sized blocks are utilized for making connections to two-wire signals. For instance, one block can be used for an RS-

485 signal (A/B) whereas another block for an external sensor power supply. Again, screw terminals offer a secure screw clamp connection and simplify wiring.



Fig 7.2 2-pin PCB Mount Terminal Block

- **Berg Stick (Female) Headers** – They are female pin headers soldered onto the PCB. Such female headers offer sockets for plug-in connection of external wires or modules. For example, one female header on the PCB can be marked as "A0, GND, +5V," which indicates the female socket can be used to connect both the analog signal (A0) and power/grounding connections for a sensor. Thus, female headers allow for plugging wires with male pins to make quick connections without any soldering. The board contains two female Berg Headers for separate signals.



Fig 7.3 Berg Stick (Female) Headers

- **Berg Stick (Male) Header** – It is a male pin-header (pins upwards) providing a socket connector for external connection points. They can be used to make connections between a female jumper wires and thus connect two PCB boards together. Male Berg Headers provide connection points allowing the board to communicate with other circuitries, much like an Arduino pin plugged into an appropriate socket.



Fig 7.4 Berg Stick (Male) Headers

- **100 μ F Capacitor** – Large electrolytic capacitors (100 μ F) are mounted close to the voltage regulator input (input side), acting as a filter for incoming power fluctuations. The purpose of using such capacitor is to stabilize the incoming power before feeding it to the regulator. The input capacitor buffers the voltage to smooth out abrupt changes (i.e., sudden drops when loading occurs) in voltage supplied.



Fig 7.5 100 μ F Capacitor

- **47 μ F Capacitor** – This electrolytic capacitor sits on the output side of the regulator (between +5 V and ground). It filters the regulator's output, reducing ripple and transient noise on the 5 V line. In combination with the input capacitor, it ensures the 5 V supply is clean and stable under varying loads. (In fact, using capacitors on both the input and output of an LM7805 regulator is standard practice to stabilize the voltage and reduce noise).



Fig 7.6 47 μ F Capacitor

- **LM7805 Voltage Regulator with Heatsink** – The LM7805 is a fixed +5 V linear regulator IC. It takes the higher input voltage (e.g. +12 V from the battery or adapter) and outputs a steady +5 V for the system's electronics. This 5 V output powers the microcontroller and many sensor modules. The LM7805 has built-in features (short-circuit protection, thermal shutdown) to protect the circuit. Because it dissipates the voltage drop as heat, we attach a small heatsink to it on the PCB to keep it cool under load.

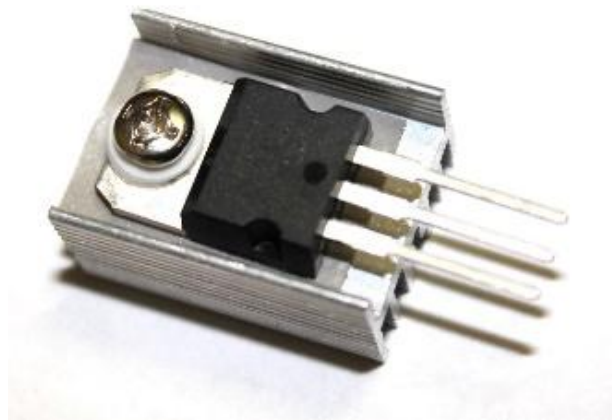


Fig 7.7 LM7805 Voltage Regulator with Heatsink

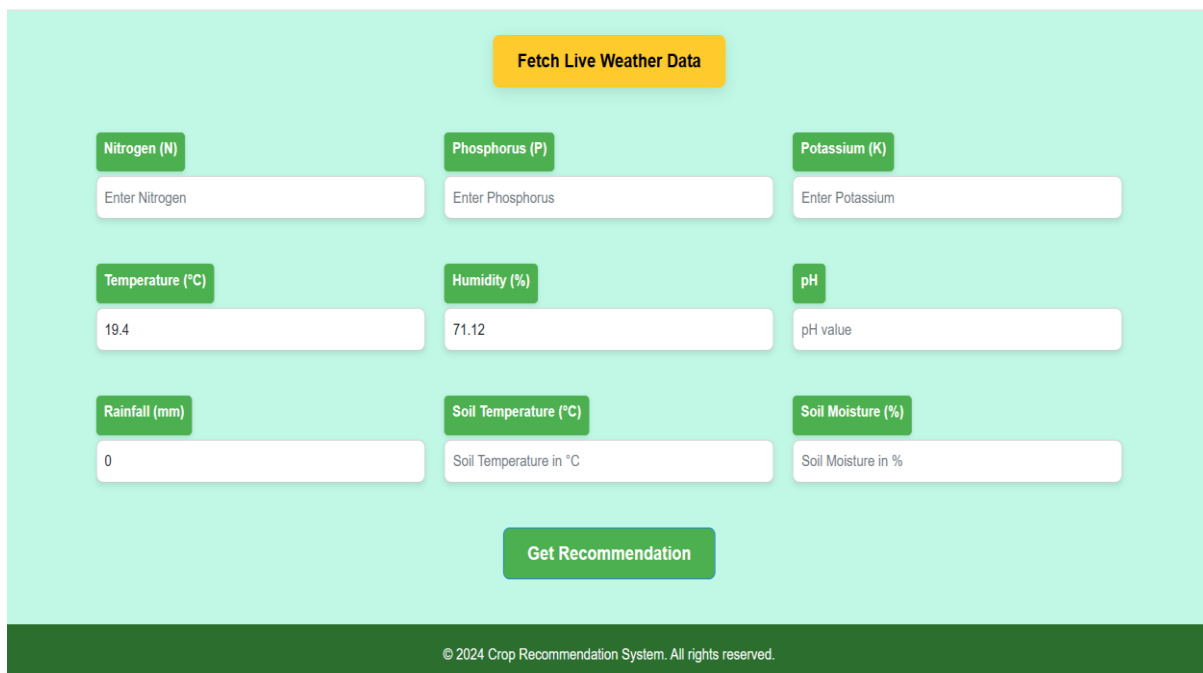
Each of these PCB components replaces part of the old breadboard setup: the terminal blocks and headers provide organized connection points, and the capacitors/regulator furnish a stable power supply. Altogether, the PCB implementation yields a neat, reliable wiring scheme that improves on the temporary breadboard wiring in every way.

7. RESULTS AND DISCUSSION:

7.1 Software Output:

The software module of the Crop Recommendation System successfully processes real-time and user-entered agricultural parameters through the web interface. The interface (Fig 9.1) collects values such as nitrogen (N), phosphorus (P), potassium (K), temperature, humidity, rainfall, soil temperature, soil moisture, and pH. Once the user clicks on “Get Recommendation”, the backend machine learning model analyses the input data and generates the most suitable crop for cultivation. It also fetches live weather data and fills the Rainfall, Humidity and Temperature section itself based on the temperature received through the API.

The predicted result is then displayed on the interface along with a visual representation (Fig 9.3) and a short descriptive message, for example: “Banana is the best crop to be cultivated here.” This output confirms the correct functioning of the integrated frontend, backend, and ML model, demonstrating the system’s capability to provide accurate, data-driven crop recommendations based on environmental and soil conditions.



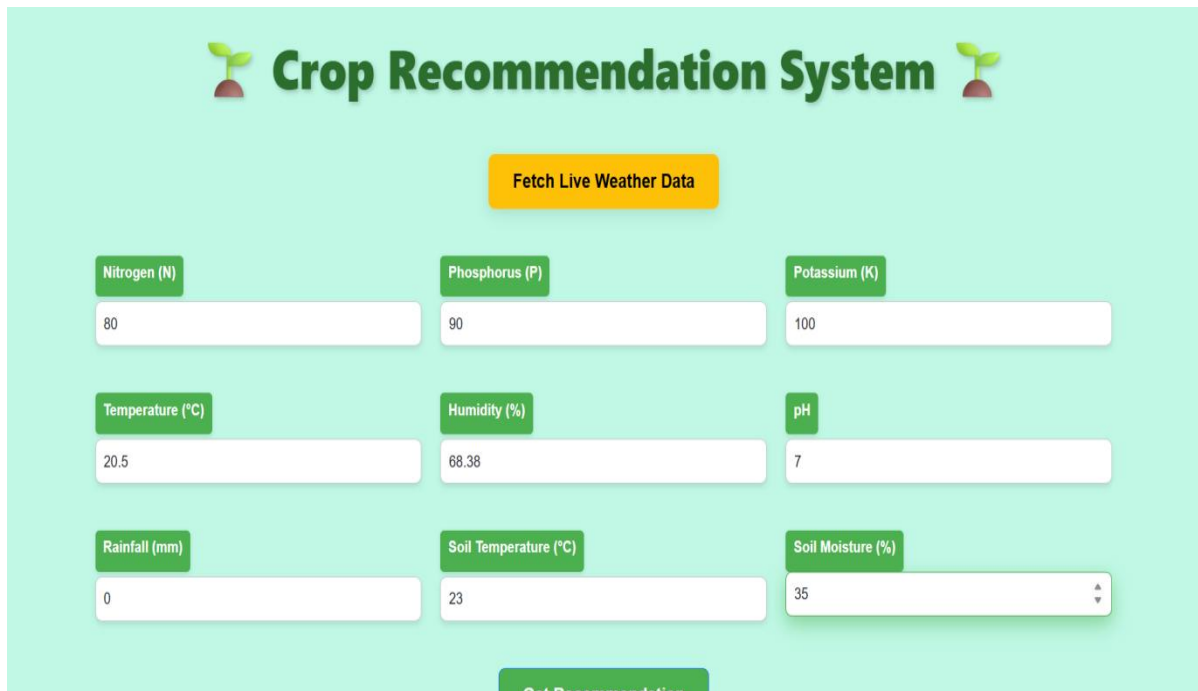
The screenshot displays the web interface of the Crop Recommendation System. At the top, there is a yellow button labeled "Fetch Live Weather Data". Below this, the interface is organized into a grid of input fields for various agricultural parameters:

- Nitrogen (N):** A green header with a white input field labeled "Enter Nitrogen".
- Phosphorus (P):** A green header with a white input field labeled "Enter Phosphorus".
- Potassium (K):** A green header with a white input field labeled "Enter Potassium".
- Temperature (°C):** A green header with a white input field containing the value "19.4".
- Humidity (%):** A green header with a white input field containing the value "71.12".
- pH:** A green header with a white input field labeled "pH value".
- Rainfall (mm):** A green header with a white input field containing the value "0".
- Soil Temperature (°C):** A green header with a white input field labeled "Soil Temperature in °C".
- Soil Moisture (%):** A green header with a white input field labeled "Soil Moisture in %".

At the bottom center, there is a green button labeled "Get Recommendation". The footer of the page contains the text "© 2024 Crop Recommendation System. All rights reserved."

Fig 8.1 Web Page View 1

Input interface of the Crop Recommendation System displaying fields for entering soil nutrients (N, P, K), temperature, humidity, pH, rainfall, soil temperature, and soil moisture.



The image shows a web page titled "Crop Recommendation System" with a light green background. At the top, there are two small plant icons flanking the title. Below the title is a yellow button labeled "Fetch Live Weather Data". Underneath this button are nine input fields arranged in a 3x3 grid. Each field has a green label above it and a white input box. The labels and their corresponding values are: Nitrogen (N) with 80, Phosphorus (P) with 90, Potassium (K) with 100, Temperature (°C) with 20.5, Humidity (%) with 68.38, pH with 7, Rainfall (mm) with 0, Soil Temperature (°C) with 23, and Soil Moisture (%) with 35. At the bottom center, there is a green button labeled "Get Recommendation".

Nitrogen (N)	Phosphorus (P)	Potassium (K)
80	90	100
Temperature (°C)	Humidity (%)	pH
20.5	68.38	7
Rainfall (mm)	Soil Temperature (°C)	Soil Moisture (%)
0	23	35

Fig 8.2 Web Page View 2

Updated Crop Recommendation System form with sample values filled in, ready for processing and crop prediction. Live weather data fetched after clicking “Fetch Live Weather Data button using OpenWeatherMap APIs.

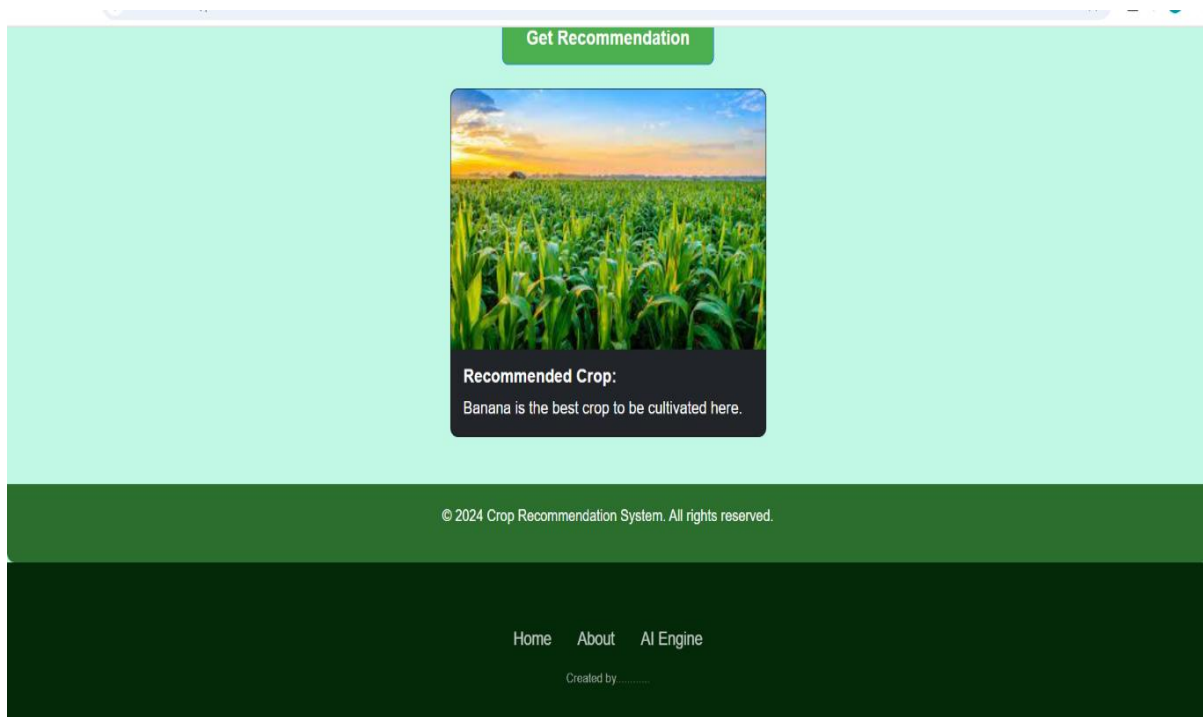


Fig 8.3 Web Page Output

Output section of the Crop Recommendation System showing the predicted best crop (Banana) based on the input soil and environmental parameters.

7.2 Hardware Output:

The output obtained from the Arduino Serial Monitor (Fig. 8.4), after sensing soil moisture and soil temperature through the connected sensors, is observed and recorded by the user as real-time soil condition data. These readings are then manually entered into the web-based interface along with other relevant parameters such as pH, NPK values, and environmental factors required for crop prediction.

Once the data is submitted, it is processed by the backend system and fed into the trained machine learning model, which analyzes the combined inputs to predict the most suitable crop for cultivation. The result is displayed on the user interface in an understandable format, enabling users to make accurate, data-driven decisions based on real-time soil and environmental conditions, thereby improving crop productivity and resource management.

Output:

<pre> ----- Potassium: 5 mg/kg Voltage: 2.453 V pH: 9.25 ----- Soil Moisture: 10% Temperature: 26.12 *C Nitrogen: 11 mg/kg Phosphorous: 4 mg/kg Potassium: 5 mg/kg Voltage: 2.458 V pH: 8.83 ----- </pre>	<pre> ----- Potassium: 6 mg/kg Voltage: 2.426 V pH: 10.25 ----- Soil Moisture: 17% Temperature: 26.12 *C Nitrogen: 12 mg/kg Phosphorous: 4 mg/kg Potassium: 6 mg/kg Voltage: 2.441 V pH: 9.72 ----- </pre>	<pre> ----- Potassium: 4 mg/kg Voltage: 2.373 V pH: 12.27 ----- Soil Moisture: 48% Temperature: 26.12 *C Nitrogen: 8 mg/kg Phosphorous: 3 mg/kg Potassium: 4 mg/kg Voltage: 2.401 V pH: 11.51 ----- </pre>
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Fig 8.4 Serial Monitor Outputs

7.3 Chatbot Output

Output Generation 1

- The chatbot interface is displayed within the AgroCare web application.
- The system shows a welcome message from the AgroCare assistant.
- A chat input box and send button are available for the user to type queries.
- At this stage, the chatbot is waiting for the user's question.

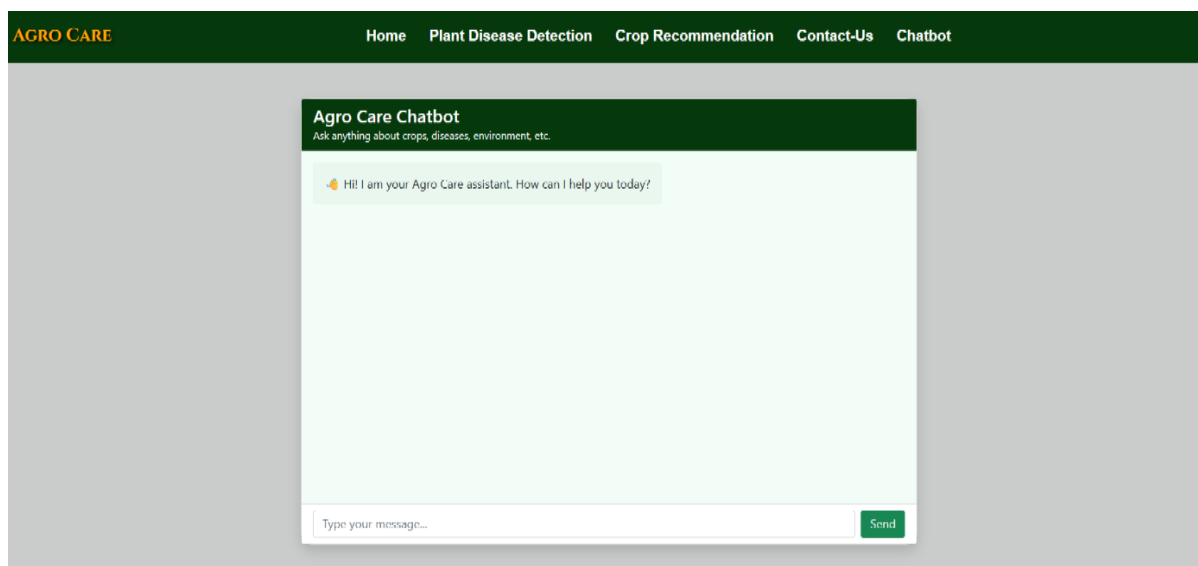


Fig 8.5- Chatbot Output generation 1

Output Generation 2

- The user submits a query asking about the climate conditions required for growing mangoes.
- The chatbot processes the query and generates a structured response.
- The response includes agricultural information such as temperature, rainfall, humidity, and sunlight requirements.
- The chatbot displays the answer in a formatted bullet-point structure for better readability.

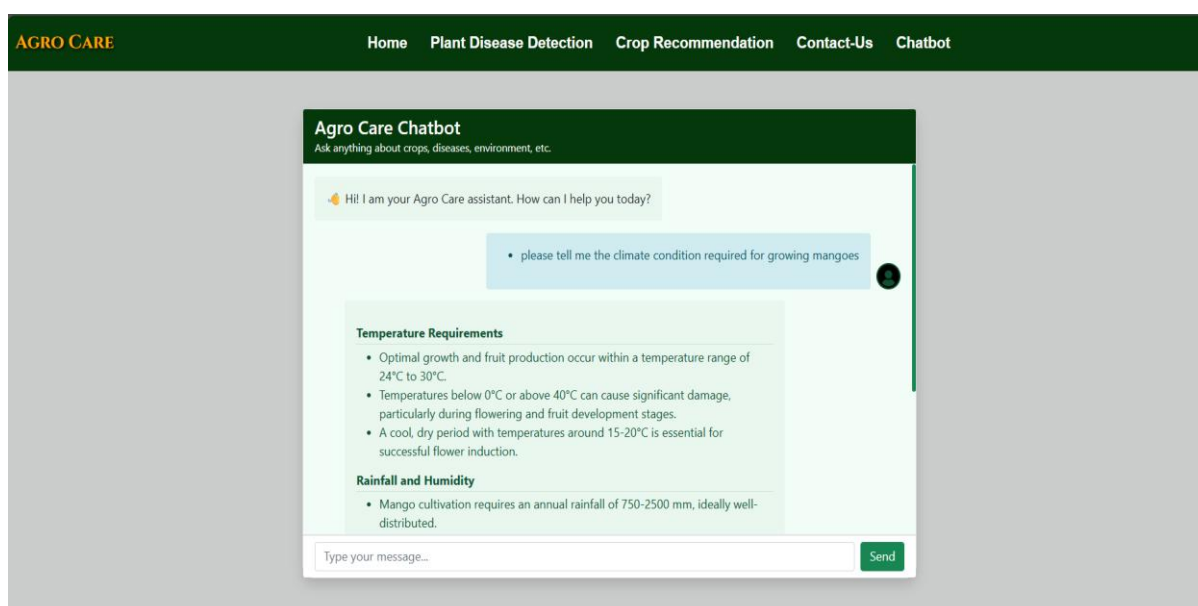


Fig 8.6-Chatbot Output generation 2

Output Generation 3

- The chatbot continues displaying the complete generated response with detailed climate conditions.
- The chat window maintains the conversation history, where both the user query and chatbot response remain visible in the chat interface.
- The system stores previous messages in session-based chat history, allowing the chatbot to remember earlier interactions and respond correctly to follow-up questions.
- This ensures continuous conversation flow and contextual understanding during the interaction.

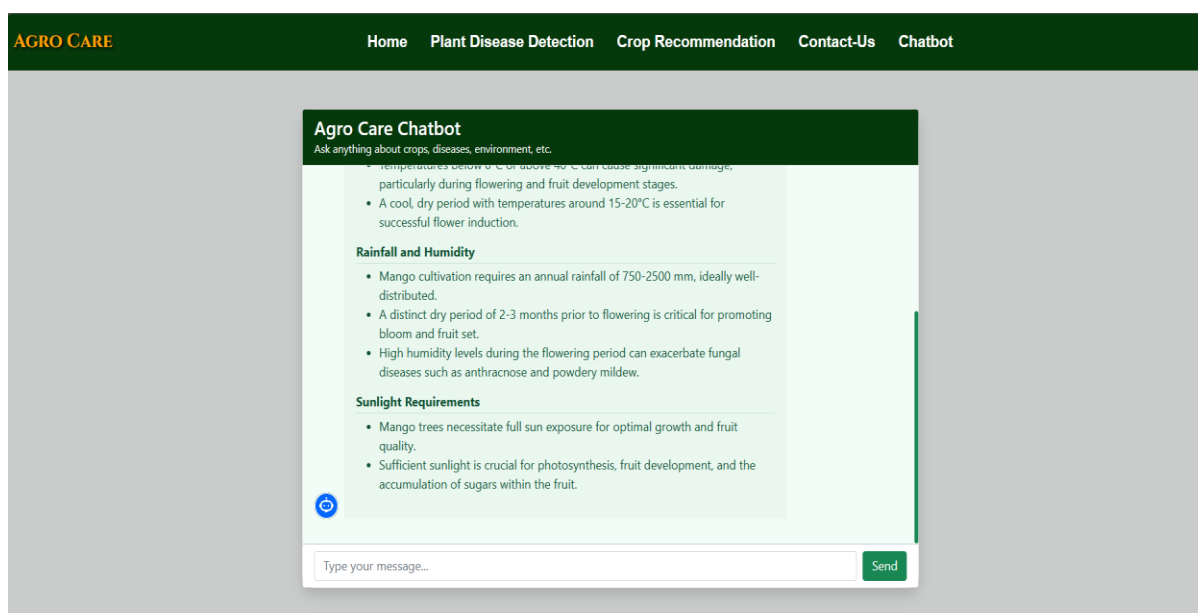


Fig 8.7- Chatbot Output generation 3

8. CONCLUSION

The Smart Crop Prediction System successfully integrates IoT sensors, real-time weather data, and machine learning to provide an efficient and reliable solution for modern agriculture. By collecting soil moisture, soil temperature, environmental factors, and essential nutrient values, the system creates a comprehensive dataset that enables accurate crop recommendations. The seamless flow between hardware, frontend interface, API services, and the ML model ensures that users receive actionable insights instantly.

This project demonstrates how technology can significantly support farmers in making data-driven decisions, improving crop yield, reducing risk, and promoting sustainable farming practices. It highlights the potential of combining IoT and AI to solve real-world agricultural challenges and serves as a strong foundation for future enhancements such as automated irrigation, real-time dashboards, and mobile app integration.

9. FUTURE SCOPE

In future, the system can be enhanced by enabling complete automation of data acquisition and transmission, eliminating the need for manual input of sensor values into the web interface. This can be achieved by integrating a Wi-Fi-enabled embedded module capable of both wireless communication and location tracking through an inbuilt GPS unit. Such a module would continuously collect real-time soil and environmental parameters and transmit them directly to the server.

To automate the data flow, the system can utilize IoT middleware platforms (such as Spider or similar automation frameworks) that act as a bridge between hardware and cloud services. These platforms can handle real-time data streaming, device communication, and API-based integration with the backend server. Sensor data can be automatically pushed to the cloud-hosted application without human intervention.

The web application, deployed on a cloud server, can be connected to this automated pipeline through RESTful APIs. As soon as the data is received, the machine learning model processes it and generates crop recommendations in real time. This creates a fully autonomous system where data collection, transmission, processing, and output generation occur seamlessly.

Additionally, the inclusion of GPS capabilities allows the system to provide location-specific recommendations, improving the accuracy of predictions by correlating environmental data

with geographical conditions.

Further improvements can be made in terms of power efficiency by incorporating advanced power management techniques, such as solar energy harvesting, low-power operation modes, and intelligent sensor scheduling. These enhancements will make the system more suitable for long-term deployment in remote agricultural fields.

Overall, these advancements will transform the current system into a fully automated, location-aware, and scalable smart agriculture solution, capable of operating efficiently in real-world conditions.

10. REFERENCES

1. Ferentinos, D.P. (2018). Deep learning models for plant disease detection. Computers and Electronics in Agriculture.
Link- <https://www.sciencedirect.com/science/article/abs/pii/S0168169917311742>
2. Patil, A. et al. (2019). Plant disease detection using machine learning techniques.
Link- <https://ijai.iaescore.com/index.php/IJAI/article/view/25709>
3. Singh, V. et al. (2020). Crop disease classification using Random Forest.
Link-
https://www.researchgate.net/publication/397121468_Soil_Based_Crop_Recommendation_System_Using_Machine_Learning
4. Rajeswari, S. et al. (2022). IoT-based smart agriculture monitoring system.
Link-<https://www.iosrjournals.org/iosr-jce/papers/Conf.2023/Volume%202/1%2C%2001-04.pdf>
5. <https://docs.arduino.cc/hardware/nano/>
6. <https://cdn.sparkfun.com/datasheets/Sensors/Temp/DS18B20.pdf>
7. <https://5.imimg.com/data5/SELLER/Doc/2022/6/XB/EU/YX/5551405/soil-sensor-jxbs-3001-npk-rs.pdf>
8. <https://www.supmeaauto.com/uploads/2101/ph-sensor-datasheet.pdf>
9. <https://robocraze.com/blogs/post/all-about-ph-sensor?srsltid=AfmBOorTQkWE4tpgUkLyEZUjiIf0qxi352RUwtjjFt3dy-tNsjd7Tlx>
10. https://www.onsetcomp.com/sites/default/files/resources-documents/15081-I%20MAN-S-SM.pdf?srsltid=AfmBOorRlzU_WSL5jAuwn1MVEuPfiaRJxaoOslBqMMnX9e8tEDdBi3vO
11. https://www.researchgate.net/publication/371071895_IOT_Based_Smart_Agriculture_Monitoring_System
12. <https://how2electronics.com/iot-based-soil-nutrient-monitoring-with-arduino-esp32/>
13. <https://ieeexplore.ieee.org/document/10212196/references#references>